

The history of alternative fuels in transportation: The case of electric and hybrid cars

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Abstract

The article describes and presents a critical analysis of the long history of alternative fuels and propulsion technologies, particularly in automobile applications. Cases are electric and hybrid cars. The term “critical analysis” refers to the analysis of the various alternative technologies in relation to their societal contexts. In particular, these are the varying contexts of energy security, energy policy, environmental problems, sustainability, and also the later more explicit climate change context. This approach gives some knowledge with relevance to the current discussions on implementation issues. The work is first of all founded on the knowledge field of “Social Studies of Technological Systems”. © 2007 Elsevier Ltd. All rights reserved.

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1. Electric cars—the early innovators

The history of electric cars is closely related to the history of batteries (Wakefield, 1994; Sperling, 1995; Westbrook, 2001; Anderson and Anderson, 2005). The names of all the early innovators are still used today. In 1800, the Italian Alessandro Volta demonstrated that electric energy could be stored chemically. He himself was inspired by some earlier experiments made by his countryman Luigi Galvani. Galvani was a professor in medicine, who carried out some rather cruel experiments in which he observed the twitching of a frog's leg when subjecting it to what would later be known as electric current. In 1821, the Briton Michael Faraday demonstrated the principles of the electric motor—or generator—while applying Volta's chemical pile as a component of his experiments. Later, in 1831, Faraday showed the principles of electromagnetic induction with the close relation between electric currents and magnetism, thereby laying the foundation for the electric motors and generators explicitly required for electric cars. The first experimental light-weight electric

vehicles thus appeared both in the USA, UK and the Netherlands in the mid-1830s. Extensive developments in electrochemistry took place in these early years. In 1859, the Belgian Gaston Planté made a path-breaking demonstration of the first lead-acid battery cell. This implied the invention of the lead-acid battery still used as a starter battery in all internal combustion engine (ICE) cars and also as a power battery in most electric cars. Other chemical cell batteries were further developed in these years, for instance the iron–zinc battery. And only a few years later, in 1861, another Italian, Antonio Pacinotti, invented the “ring” direct current motor. The first electric vehicle—a tricycle—applying the Planté lead battery as a power source was demonstrated in France by a Mr Trouvé in 1881. In the Seine the same year, he actually demonstrated the first electric boat with a similar power source. During these years, the early 1880s, other similar electric tricycles with lead batteries were also demonstrated in the USA and UK. In this context, it is worth remembering that the German Benz demonstrated the first ICE vehicle in 1885.

Later on Thomas Edison joined forces, as he saw great opportunities for electric cars. He made substantial efforts to develop more efficient batteries. In 1901, he came up with the nickel–iron battery, which was very much in focus when

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electric car interests highlighted again in the ensuing century. The nickel-iron battery could store 40% more energy per weight unit than the lead battery, but its production costs were very high. The cost problems were so extensive that the battery was precluded from a wider use in commercial electric cars. Both the nickel–zinc and also the quite advanced zinc–air battery were invented during the same period. It is even assumed that the zinc-air battery was actually applied to an electric vehicle before the turn of the century.

2. The golden age

The couple of decades that followed, from 1880 to around 1900, should become the most intensive period in electric car deployment. It represented the beginning of *the golden age*, ending again in the early 1920s (Westbrook, 2001). Electric cars have never again experienced any similar prosperous age of technological development and deployment. Most major technological breakthroughs were achieved in this period. Even now, a hundred years later, they still form the basis for electric car technologies (Anderson and Anderson, 2005).

The 1893 World Exhibition in Chicago featured six types of electric cars. Commercial fleets were early deemed necessary to achieve a wider application of the new cars. Taxi fleets in major cities were obvious candidates. The taxi companies maintained the batteries in their common garages and the daily distances travelled by the “cabs” were well within the battery range. Thriving businesses were set up in London, New York and Paris before 1900. The taxi drivers themselves were reported to be very enthusiastic about these new cars, firstly called “horseless carriages” only soon later to be called “automobiles”, a term first used by a London newspaper in 1895. Already in 1897, there were 15 such taxis in London and 13 in New York.

At this time, three types of cars were contesting for market control; the electric car, the steam engine and the gasoline ICE car. In 1903, New York had about 4000 registered motor vehicles: 53% steam-powered, 27% gasoline ICEs, and 20% electric. But already in 1899–1900, the electric cars outsold the other two types in the USA as a whole. In 1912, the peak was reached in the USA with about 30,000 electric vehicles altogether. But at this point, since 1909, the Ford-T model had really begun to achieve market dominance. The electric car was the most conservative of the three, with its close resemblance to horse carriages both in appearance and performance. The manufacturers also copied the more fashionable horse carriage forms, something considered to be a marketing asset (Wakefield, 1994).

Many technological and infrastructural innovations took place in this period. The desired fast charging of the batteries was of course a challenge at that time, as it is now. Grids of charging stations were established, and there were continuous discussions about the further expansion of these. However, this new infrastructure would never be able to compete with the extensive development of gasoline stations that began more or less in the same period. The most inventive development took place in New York. Around 1900, a system of *charging hydrants* was established, i.e. a coin-operated mechanism

with both voltmeter and wattmeter. It was set up and driven by an electricity supply company. When the driver deposited the right amount of coins, he could get out the number of watt-hours needed for charging. This system was assumed to greatly enhance the use of electric vehicles, at least in city areas. But there were also ambitions to make the electric car competitive for longer, extra-urban travelling. Electric cars with easily interchangeable battery systems were for instance developed to overcome the range limits. And already from 1895, they were used quite extensively in order to achieve “touring” with electric cars over longer distances, notably in several early car races where they competed with ICE and steam engine cars both for reach and speed records. Another somewhat later invention for the same purpose was a fast battery swapping system, with batteries on rollers that allowed the car owners to easily roll out the discharged batteries and roll in the new charged ones (Anderson and Anderson, 2005).

Besides intensive efforts to develop new battery types and inventive systems for fast battery exchange and recharging, two other technologies were developed around 1900 which aimed at enhancing the reach of the electric car. Firstly, there was the principle of *regenerative braking*. This was demonstrated in Paris in 1887, utilising the ability of the electric drive motor to act as a generator charging the battery when overdriven mechanically by the vehicle wheels. In this way, driving downhill gives extra power back to the battery. The recharging loads the drive motor—as a generator—enough both to achieve the necessary braking effect and to increase the energy stored in the battery. Early estimates showed that one would expect increases in car ranges up to 40%, just as the estimates would be some hundred years later. Such regenerative braking became standard equipment in many of the models already sold in the early years of the 18th Century (Wakefield, 1994; Westbrook, 2001).

The second important technology developed to increase the limited reach of the battery cars was *the hybrid*, so much more in focus at that point than a hundred years later. The notable car developer Ferdinand Porsche was one of the first inventors in this field. His gasoline–electric car was shown at the Paris Exposition in 1900. These early hybrids also included regenerative braking technologies. Such a gasoline–electric hybrid car was demonstrated as early as the Third Annual Automobile and Cycle Show in Paris in 1901. The American Electric Vehicle Co. in Columbia announced in 1902 in an advertisement the sale of its electric–gasoline hybrid, “to afford the maximum of safety, reliability, comfort and luxury (Write for catalogue of seventeen different Columbia models)” (Anderson and Anderson, 2005). The hybrids were claimed to combine the best of two technologies; noiseless when driving in city areas and with no limits to the reach of the car outside the cities. The American journal *The Automobile* stated in a special issue in 1905 “...it certainly seems that there is a big future before the petrol-electric car” (Anderson and Anderson, 2005). Mostly due to cost problems, the hybrid concept did, however, more or less disappear during the First World War, never to be looked seriously at again before the early 1970s (Wakefield, 1998).

3. Wartime conditions

At this time, however, the electric vehicles still constituted a thriving industry. The War in Europe only increased the production and development potentials. In the USA as a major world exporter, and in England, Germany and France due to the fact that most gasoline vehicles were requisitioned the war effort. Whole municipal fleets of electric vehicles were used for street cleaning and garbage collection, and many merchants had their own private fleets of electric vans for retail and goods deliveries. England for instance had more than 1000 electric trucks in these years. The infrastructure was also developed. Birmingham district is known to have had seven charging stations only in the city area, and more than twenty in the surrounding district. Both in the USA and Europe in this period, large efforts were put into the enhancement of recharging infrastructures and potentials.

Two conditions were crucial to this war peak in production and use: firstly, the gasoline shortage and the requisitioning of ICE vehicles to take part in the war effort; secondly, an extensive development of new electric power stations took place—mostly large coal-fired power stations as known from England and Germany, but also large hydro power stations in Italy, Norway and Sweden. Electricity seemed to become an abundant resource, available not only for industrial purposes but for transportation as well.

By the end of the First World War, the USA alone had an estimated 50,000 electric vehicles altogether, and they were exporting large numbers to war-torn Europe, mostly cars for private passenger transport. At an automobile show in New York in 1918, cars from four major electric vehicle companies were demonstrated. A number of buyers from many European countries including Norway and Sweden, but also Japan and South America, were there to place large orders. A Norwegian for instance immediately placed an order for one hundred cars, ready to buy another forty within a fortnight. Several electric taxis were sold to Japan (Wakefield, 1994; Westbrook, 2001; Anderson and Anderson, 2005).

Expectations were high, but they would soon fade. The electric vehicles lost ground to the gasoline and diesel ICE vehicles. Some companies were still producing various models throughout the 1920s. However, sales were relatively low, and with the stock market crash in 1929 and the international economic depression, most of the companies left went bankrupt. In the USA, electric car production did not resume again before the Second World War. Again there was a wartime peak, caused by gasoline shortage and the priority given to warfront use of both gasoline and diesel. A similar peak was experienced in most European countries, and in Japan, where the electric car production continued until the early 1950s due to prolonged gasoline shortage. German authorities in particular actively promoted the use of electric vehicles by making them tax-exempt, emphasising the fact that this by far is a new measure. During wartime they had about 30,000 electric vehicles running, for instance for postal service. Electric vehicles were thus seen on the roads in quite significant numbers, but this time to a larger extent together with ICE vehicles

applying various types of alternative fuels from local renewable resources, notably bioalcohols and biodiesels (Anderson and Anderson, 2005).

Within this wartime context, Great Britain has a somewhat different story. It is basically the story of the electric milk vans, the most long-lived fleet of electric vehicles the world has ever seen, a fleet that is still operating. In the 1950s, the fleet grew to a total of about 30,000, a number kept fairly stable in the decades to come. It was an ideal fleet for electric driving. The vans were noiseless when delivering milk early in the morning, and they could be parked in common garages for recharging during the night (Westbrook, 2001; Anderson and Anderson, 2005).

Actually, the British started their renaissance of the electric vehicle well before the war broke out, in the mid 1930s, mainly due to the availability of large amounts of cheap electricity. By 1940 they had more than 6000 electric vans running, mostly used for milk and bread deliveries. To enhance the use of electric vehicles further, British authorities had during the war made strong marketing campaigns, very similar to the advertising campaigns shown some thirty years earlier. All the advantages of electric driving were starkly emphasised; they had a long life and would help to conserve natural resources; they were simple and cheap to operate; they required less maintenance, and hills were not a problem as they made recharging possible when driving downhill. The British were certainly striving to compete with the Germans in promoting electric driving. When it came to long life it was, however, not a success. Setting aside the milk van story, the production and use of electric cars would more or less completely fade out again during the first post-war years (Anderson and Anderson, 2005).

In Japan, the post-war period was of course rather problematic with severe shortages and rationing of gasoline. This initiated the production of an electric car—Tama Electric Powercar—which would become quite popular. It had an impressive range of about 150 km and a driving speed of almost 60 km/h. But when gasoline became readily available in 1952, the production company turned to ICE cars. From this period, we also know of several hybrid gasoline—electric cars, both in France and the USA. In the case of the USA, even a hybrid sports car was produced, but only for a few years (Wakefield, 1994).

4. From Silent Spring to silent cars

In the 1960s, a quite new debate turned up. Rachel Carson (1963) published her book *Silent Spring* in 1962. This book is by many considered to represent the real advent of the modern environmental debate, as we have experienced it since then. Carson focused on environmental pollution problems, mostly caused by pesticide chemicals used in agriculture. But in parallel, a separate debate took place on air pollution problems in larger cities. It comprised issues such as lead pollution, emissions of fine particulates, carbon monoxide and nitrogen oxides, and smog creation. These were the years when lead was still an additive in all gasoline, and when there were no

particle filters or catalysts. In all cases, the focus was limited to purely local contexts. Environmental problems were seen as local problems, which could also be solved locally. The ICE private car was identified as one of the large sinners, and the electric car as a solution.

Now, for the first time the world major car producers showed interest in electric cars. In 1966, the Research Staff at the British Ford Motor Company were asked to construct a small urban electric car, responding to the following requirements (Westbrook, 2001):

1. Small enough to occupy minimum road and parking space
2. High manoeuvrability
3. Minimum pollution
4. Simple to operate
5. Low initial and running costs

Comuta was the name given to the 1967 prototype. A name with obvious connotations to urban *commuting*, a phenomenon in large growth at that time. With lead batteries it had a maximum range of about 60 km at a speed of 40 km/h, and a maximum speed of somewhat above 60 km/h. Not very impressive figures compared to the models produced some sixty years before. Even though it had a heating system with recycled waste heat from the motors, it was a general perception that the car was not suitable for use under winter conditions. Fairly soon the conclusion would be that the *Comuta* was not promising enough to be developed for commercialisation. One of the prototypes is currently on display at the London Science Museum (Westbrook, 2001).

In the mid-1960s, US General Motors initiated an electric car development programme. The prototypes *Electrovair* and *Electrovan* seemed quite advanced. They had a three-phase AC drive system with power either from a silver–zinc battery or a fuel cell. In this period, GM also converted an Opel Kadette to electric driving, using a zinc–air battery with a claimed range of more than 200 km. But it never came any further than prototype production. Generally, the 1960s only demonstrated the difficulties in developing electric cars with acceptable ranges, driving performances and, not least, costs (Westbrook, 2001).

5. Soft energy paths

In the 1970s, all over the Western World, energy problems would very soon become an integrative part of environmental discourse. Three international events would become particularly influential on the further development of the discourse itself as well as the more substantive processes. In 1972, The Club of Rome published the book *The Limits to Growth* (Meadows et al., 1972). It highlighted the issue of absolute global limits to future growth in the exploitation of non-renewable natural resources, among others fossil energy resources. Later this type of understanding has been connected to the term *limits of sources*, as a distinct category from *limits of sinks*. Limits to global emissions of CO₂ represent an example of the last category. As a second event much of the

Western World was hit by an oil crisis in 1973—in the public debate even called an energy crisis, but actually caused by an oil embargo effectuated by the major Arabic oil producers. The result was quite extensive rationing of oil use, for instance in Norway with prohibition against the use of private cars during weekends. Highways without cars, but with people skiing on them became a public symbol of the crisis. The third event—the nuclear power debate—was more like a course of events. With a large degree of intensity it took place again from the early 1970s in almost every Western World country.

All three events served to bring into focus the need to develop alternative, renewable energy resources and technologies. Within very few years, solar, wind, wave, bio-energy, and heat pumps based on natural heat resources became integral parts not only of the public energy debate, but also of governmental energy development plans. Energy scenarios—founded on the use of renewable resources—were drawn up in almost every country, and substantial efforts were put into research and development (R&D). A prominent American contributor at that time—Amory Lovins (1977)—summarised this in his book entitled *Soft Energy Paths*, with the sub-title *Toward a Durable Peace*.

The 1970s would become a very active period in the history of electric car development. These cars were seen to be part of the soft energy paths, by combining zero polluting emissions with the possibilities of utilising solar cells, wind and wave energy as sources of the electric energy needed. In the USA, as well as Europe and Japan, most major car producers were involved in developing electric cars with various drive systems and battery types. But this development never really took off. A new company was set up in the UK to produce electric cars solely for a niche market of specialist electric vehicles. However, only 300 vehicles were actually built, and the company was closed down in 1979. And at that time, the activities in electric car development once more faded out worldwide (Wakefield, 1994).

6. The French VEL electric car and reverse salients

As a case story from this period we shall turn to France, where large efforts were put into a major innovation process in the 1970s: the electric car called VEL. It is a case story described and analysed in several works by Michel Callon (1980, 1999). This presentation is based on those works. Callon has used the case in his contributions to actor network theory in innovation studies. It is worth noting that not only was the VEL a battery electric car, but it was also envisaged to develop into a hydrogen fuel cell car. The real large markets of private transport were first of all considered to be obtainable through breakthroughs in fuel cell technology and with fuel cell cars that could achieve speeds up to 90 km/h and have a reach larger than conventional battery electric cars. Such a development in new basic energy conversion technologies was described as realistic and obtainable within the late 1980s.

It was a group of engineers at the French electric utility company that initiated the VEL project. The project was ambitious. Not only did they presuppose that the technical

problems could be overcome quite fast, but also radical changes in the French social structures. They acted as engineer-sociologists, a term used by Callon. As sociologists they were inspired by the student revolt initiated in France in May 1968, and the ensuing discussions on environmental problems and a post-industrial society with strong social movements and anti-consumerism values. The traditional motor car—with combustion engines—was linked to the “old” industrial society and the air pollution and noise that plagued the cities. However, it was also linked to the consumer society in which the private car constituted a primordial element of status. They *re-defined* the car, to use a term that has been applied in the later discussions in the 1990s on the future of the electric car. Of course, the VEL car was to be non-polluting and silent. But it was not least seen as an ordinary means of transport, without connotations to speed, acceleration, freedom, looks, and social status. A new society was conjectured in which people had changed their values and their relationships regarding the car. Included there were fundamental changes in urban planning and policies. All crucial urban actors were assumed to cooperate around a prime focus on transport and environment and development of comprehensive electrified transport systems, both public and private.

Similarly, a re-definition of the whole production structure of cars was required. New roles for various major industrial companies were assigned and defined in detail by the VEL engineer-sociologists. The major French electricity company was supposed to develop the electric motor, to perfect the lead batteries that would be used in the first generation of VEL cars, and to carry out the development of second generation zinc–air batteries. Renault would, on the other hand, have a much more limited role; from the manufacturing of complete cars to only making car bodies and chassis assembling. It could no longer regain the status as one of the most powerful European car manufacturers and industrial companies. It would be a major change in industrial structure, with the electric industrial complex as the backbone and strongly reduced importance of the formerly dominating oil and car industrial complex.

The VEL project never really left the laboratories and the drawing boards. Things would soon begin to go wrong for the responsible engineers. Resistance appeared fairly spontaneously and unexpectedly in several places, much like in guerrilla warfare according to Callon. *Reverse salients* is the term applied by Thomas P. Hughes (1983, 1999) to describe such resistance. This is also a term he borrows from warfare terminology. In warfare, a salient is a forward wedge driven into the enemy's battle front. According to the analyses by Hughes, reverse salients develop as technological systems expand. They are components in the system that have fallen behind or are out of phase with the others. Due to the uneven and complex change it suggests, he prefers it to the more visual concept of bottleneck, a concept more widely applied in transport system analysis and theory.

The first reverse salients turned up in the basic technical components. VEL cars would need batteries with performance adapted to the requirements of average car users. However,

such batteries were not available and would be too expensive to produce for many years to come. Also, the second generation zinc–air batteries soon proved to be a shaky venture only elaborated by a few researchers at the national electricity company. But not least did the research on fuel cells experience serious setbacks; new cheaper types of catalysts that were considered necessary had in laboratories the tendency of quickly becoming contaminated, thus making the fuel cell unusable.

In his analysis, Callon emphasises the fact that the conventional motor car was completely rehabilitated through an alliance between Renault engineers, fuel cell catalysts that did not function, and a fading social protest movement. Now it was the turn of the VEL engineers to become quiet; their project had lost all credibility and strength (Callon, 1999, p. 91).

7. Towards sustainable mobility with electric driving

By the early 1980s, electric cars would come into focus again (Ulvönnäs, 1983; Mader and Bevilacqua, 1989; Wakefield, 1994). The problems of air pollution in larger cities were once again highlighted. And by the end of the decade, in 1990, California introduced their first zero-emission regulations, which in particular sparked new initiatives in developing electric vehicles during the 1990s. *Zero-emission vehicles* became a new term applied. In this context, focus shall, however, be on quite another discourse; the sustainable development discourse. Since the late 1980s, we have experienced a new and intensified focus on the need to develop alternative energy resources and technologies in the transport sector. Not only has this focus become an integral part of the current environmental and alternative energy discourse, it has even come to dominate this discourse. The term is no longer soft energy paths, but *sustainable energy systems*, with energy for mobility purposes as a crucial part. Issues related to development of alternative energy in the transport sector have thus been integrated into the much wider discourse on *sustainable development*.

Such a link was, however, not at all evident. The volume problems caused by transport were not a topic in the report from 1987 on sustainable development entitled *Our Common Future*, as it was produced by the UN World Commission on Environment and Development (WCED, 1987). Neither were they highlighted in the common global action plan, Agenda 21, from the follow-up World Summit in Rio in 1992. To the extent that transport-related problems were discussed *directly*, they were considered as traditional problems of intensity; too many cars at one place, and first of all in the fast growing mega-cities of developing countries. But the *indirect* relations would soon begin to create a pressure. The WCED report itself emphasised the importance of environmental and climate problems caused by extensive use of fossil energy, and the need both to reduce energy use in rich countries and to achieve a substantial transition to renewable forms of energy in order to solve these problems. This report—and the major follow-up Rio-conference—did also represent the very basis for new global initiatives on the needs to reduce climate gas emissions in general and CO₂ emissions in particular; with the Kyoto Protocol as a first major result. Transport is of

course the main societal sector most fundamentally linked to fossil energy. When we look at the long historical lines, the *fossil society* and the *mobile society* have grown like Siamese twins. Decoupling is in this context a particularly challenging issue as they have grown together for such a long time, more than a century.

The volume growth in transport has been very stark indeed. While many countries in the post-industrial era have managed to stabilise or even reduce energy use in stationary sectors, energy use for transport has continued to grow. Paradoxically, this is the case not only for passenger transport but also for freight transport. This is the basis for introducing an important change in the understanding of transport problems; from *problems of intensity*—too much local pollution in urban areas—in the 1960s and 1970s, to *problems of volume* from around 1990 and onwards; too much motorised transport in general and too much energy consumption and macro-regional/global pollution in particular.

As expressions of this understanding two entirely new concepts were launched; *sustainable transport* and *sustainable mobility*. Both were soon extensively applied to research as well as to public policy contexts. In 1992, the EU termed its common transport policy “a strategy for sustainable mobility”, a term still applied (EU COM, 1992). When the basic concept is *mobility*, the broader societal patterns and volumes of movement, whether of persons or freight, are highlighted. With *transport* the focus is more limited, mainly to the physical requirements of transport means and the infrastructures and transport systems of which they form part. However, in both cases—sustainable mobility and sustainable transport—electric cars were seen as one of the major conditions for the achievement of both urban and global sustainability.

8. Hybrid cars on the move

I have briefly touched on the history of hybrid cars. It is almost as long as the history of electric cars (Westbrook, 2001) and actually, the two have been tightly intertwined. Through a history of more than hundred years, various types of hybrids have been developed. However, in this context, we refer explicitly to hybrid *electric* cars, often abbreviated HEVs. To our term electric cars with this sort of terminology we may apply the term BEVs—battery only electric vehicles. In HEV cars, the electric systems—both the batteries and the electric motor—still fulfil crucial functions, but only in combination with an internal combustion engine, either driven by petrol or diesel. Several models are now on the market. Best known are the two Japanese models: *Toyota Prius* and *Honda Insight*, which emphasise the fact that this is not a matter of backyard production. The numbers sold have become quite impressive.

The hybrids of today are produced by the major world car manufacturers of petrol and diesel cars. They are mostly produced to respond to the environmental problems caused by ICE engines; emissions of climate gases, and air pollution and noise in urban areas. In the early days, however, the hybrids were produced to combat the limitations experienced by electric cars; limited range, low battery efficiency and

charging difficulties. And they were produced by electric car manufacturers, many of them being very small companies even seen in relation to the standards of those days. Some were only backyard productions.

Possible combinations between the combustion engine and the electric drive system are almost infinite (Westbrook, 2001). At one end, a large combustion engine may be applied with sufficient power to propel the car under most driving conditions, only with a small auxiliary electric drive system needed during high acceleration or steep hill climbing. At the other end, a main electric drive system may be applied, only with a small auxiliary combustion engine with some supplementary power and with the possibility of recharging batteries. With the combustion engine as the point of departure, we may thus talk about *mild hybrids* (“mybrids”) on the one hand, and *strong hybrids* on the other.

There are, however, two major basic HEV configurations; the *series hybrid* and the *parallel hybrid*. Both were fully developed and deployed in cars introduced on the market more than a hundred years ago (Wakefield, 1998). In a *series* hybrid, the transmission of power from both the combustion engine and the electric drive system is *primarily electric*. The mechanical output from the combustion engine is used for generating electric power, and this is combined with the electric power from the batteries in an electric controller arrangement. Power can flow either way between the battery and the drive motor. While braking, the motor can then act as a generator, feeding brake energy back into the battery. In a *parallel* hybrid, the transmission of power from both the combustion engine and the electric drive system is *primarily mechanical*. The two motor systems can be applied in combination or independently. In a common parallel type, the combustion engine drives the wheels directly as in an ordinary car, but with supplementary power and regenerative braking supplied by the batteries and the electric generator system. The series and parallel hybrids may also be combined, in a series–parallel configuration. Current *plug-in hybrids*, described below, are mostly such combinations. In all major configurations, the electric drive system fulfils five functions: to start the engine, to boost engine power, to absorb brake energy, to charge batteries, and to supply car auxiliaries (Jefferson and Barnard, 2002; Westbrook, 2001). The combustion engine can in any case be downsized by up to 60%, depending, however, on the car type, battery efficiency and storage capacity (Jefferson and Barnard, 2002).

Large downsizing of the combustion engine will always meet the problems of large, heavy and costly battery packages. This is a basic problem also met by the so-called plug-in hybrids, an adapted hybrid technology highly in focus today, particularly in the USA. Several of the major car producers—notably Toyota and General Motors—have announced that such cars will be manufactured. Most current plug-in hybrids are, however, backyard reconstructions of Toyota Prius models—in some cases presented as “do-it-yourself” tool sets for sale. The battery size and capacity are increased, and the car is furnished with opportunities for battery charging from an ordinary household electric outlet. It has also been

claimed that the car with its increased battery capacity may function as a source of emergency home power in the event of an electrical grid failure. The charging may take place during the night, and the car may be driven only by electricity during daily urban driving. In this way, the combustion engine may be used less, and the fuel consumption may be reduced together with local air pollution and climate gas emissions. But basically this only brings the hybrid closer to its historical origin: the all electric—BEV—car. This is by far a new invention. In reality, many of the early hybrids were such plug-in hybrids.

9. Hybrids—nothing really new under the sun

A hybrid produced by *Krieger* in 1903 further shows that there is nothing really new under the sun in relation to hybrids. It was a series hybrid, consisting of a small petrol engine, batteries, an electric generator, and two electric motors mounted directly on the front wheels. The power produced by the engine was then supplied electrically to the drive system, and not mechanically. It was also used for charging the batteries. The car had power steering, besides the front wheel drive. The American “Milde Electric Car Company” produced several models in the period 1901–1906. Their four-seat hybrid had about the same weight as an electric car, but had a range similar to a petrol car. The petrol engine was placed beneath the driver seat and could be used for charging the batteries. The pure electric drive system was used as supplementary power when driving uphill, or as a sole power source when the petrol engine was turned off in slow urban traffic (*Anderson and Anderson, 2005*).

In the case of electric cars, the French were particularly inventive in these early years. *Jeantaud* and *Jenatzy* were two notorious competitors in the field. Their participation in car races and their continuous competition for new speed records drove their development work from electrics to hybrids. *Jenatzy* had a ready developed parallel hybrid in 1901, with a petrol engine and an electric motor that could work together or separately. Through a dynamo, the petrol engine could be used for charging the batteries, which could also be done while driving downhill (*Wakefield, 1998*).

In the years 1910–1920, the hybrids were further developed. In 1916, the American “Woods gasoline—electric car” was introduced on the market. It was a parallel hybrid in which a small four-cylinder petrol engine was connected directly to an electric generator, which through mechanical transmission was connected to the rear wheel system. As for the *Jenatzy* hybrid, it could be driven solely as a petrol car, as an electric car, with both power sources in combination, or with the electric system in a generator modus for recharging of batteries. In marketing, emphasis was placed on the combination of unlimited range and noiseless driving in urban traffic. The Canadian “Galt Motor Company” had a series hybrid on the market in 1914, again with a small petrol engine, which was used for driving a generator for electricity generation. This electricity could fulfil two purposes, to charge the batteries or to drive the car through the electric drive system,

with electricity from the battery system as an additional power source. The car was claimed to have had a top speed of 50 km/h, and a fuel consumption of 4.0 litres/100 km (*Anderson and Anderson, 2005*). Even though the top speed is not particularly impressive from a present-day standpoint, the fuel consumption level is a better achievement than the current Toyota Prius model (*Jefferson and Barnard, 2002*). But this should also be the last hybrid production of any importance. Not surprisingly, the end of the *first hybrid era* more or less coincided with the termination of the golden age of the electric car.

Several new hybrid initiatives turned up again in the 1970s, particularly in the USA. The contexts of energy crisis, focus on renewable energies, urban air pollution, and later climate change were the same as for the electric car initiatives. Combined initiatives were presented, mostly with public authorities in key roles. In 1975, the US Energy Research and Development Administration—ERDA—initiated a government programme to advance electric and hybrid car technology. Over President Ford’s veto a year later, the Congress passed the “Electric and Hybrid Vehicle Act of 1976”, establishing a demonstration programme with the explicit aim to make the USA an all-electric car economy by the year 2000 (*Anderson and Anderson, 2005*). General Motor subsequently spent substantial amounts in electric and hybrid car development. But the further interest soon faded out. By the end of the decade, it was generally concluded that neither electric nor hybrid cars were able to compete with ICE cars.

10. The second era of hybrids

In the early 1990s, the hybrid adventure began again, this time mostly on the basis of public–private partnerships between public authorities and the car industry, so typical for this period with its new public management ideology. The Clinton administration announced in 1993 an initiative called the “Partnership for a New Generation of Vehicles”, PNGV, with the aim of developing a “clean car” with a fuel consumption level up to about 4.0 litres/100 km. After a few years and one billion dollars of spending, three prototypes with this achievement were presented. All three were hybrids. However, none ever came to any production level (*Anderson and Anderson, 2005*).

Toyota was not included in PNGV. In 1997, they launched their Prius hybrid four-door sedan model on the Japanese market. The same year, Audi was the first European manufacturer to put a hybrid, the *Duo*, on the market, after experiencing setbacks in electric car development. It did, however, end as another setback. Without becoming a commercial success, the production was terminated. In this period, most European car manufacturers put their environmentally related R&D efforts into further development of the diesel ICE car, also with promising possibilities as regards reductions of climate gas emissions.

In 1999, Honda was the first to launch a hybrid on the large American market. This was the two-door *Insight* model, an immediate success. A few years later—in 2003—Honda actually marketed its second petrol–electric hybrid, the Honda

Civic Hybrid, with the appearance and drivability as the ordinary Civic model. The Toyota Prius was ready for the broad American market in 2000. It has since then become a sales success, much larger than expected by the Toyota Company. This has later also been the case on the European market. In 2004, the Toyota *Prius II* won the Car of the Year Awards from Motor Trend Magazine and the North American Auto Show. And later the same year, the first American hybrid from this era was launched: the *Ford Escape* hybrid. It was a sport utility vehicle, thus hardly a contribution to overall emission reductions. Other major car manufacturers have followed with models in the later years.

There may, however, be reasons to question the viability of the success. In an urban driving cycle, the Toyota Prius may have a fuel consumption of 4.5 litres/100 km, and 5.0 litres/100 km in an extra-urban cycle (Jefferson and Barnard, 2002). Real life figures may be 20–40% higher. The Honda models have lower consumption levels, but are still comparable in relation to the size and type of car. Fuel savings are anyhow modest in comparison with modern efficient diesel or petrol ICE cars. Moreover, the everlasting question concerning the source of electricity still exists, as it was also critically posed in the discussions some hundred years ago. If the electricity comes from the ordinary grid dominated by nuclear and fossil power production, the total environmental gains are not at all obvious. This is also the case if the cars in question would develop into plug-in hybrids. Larger consumption during extra-urban driving may represent additional problems in societal conditions under which such driving constitutes a substantial share of the total mileage. This is for instance the case of a sparsely populated country such as Norway.

Furthermore, a short time sales success is one thing; a long-term economic success may be another. Hybrid cars are more costly to produce than the pure ICE cars. This is one of the reasons why they totally disappeared from the market before 1920, and did not manage to reappear during the 1970s.

11. The end of the gilded age of the electric car

When looking at more than 100 years of history, the 1990s has been the most intensive period in relation to both electric and hybrid car research and development. All major car producers in the world engaged in quite extensive development programmes. New efforts were put into the development of more efficient batteries. A lot of international R&D conferences were held which solely focused on electric cars. Even in Norway, such a conference was arranged every year. A car factory producing the electric *Think* car was for instance established in Norway; a country without any structure or traditions of car production. The Think car company was actually later taken over by the American Ford company. Optimism was similarly high all over the world. Electric cars with ranges and speeds comparable to ICE cars were considered to be viable and commercial options by the turn of the century. Some illustrating quotes from a 1992 Nordic El.car Conference in Oslo (Oslo Energi, 1992):

“Whether the electrical car shall become common is today not a question about technology, only a question about politics”.

“The current large efforts by the battery manufacturers are of course connected to that they foresee an enormous market being opened”.

“It is today possible—based on advanced battery technology—to realise a range of about 500 km”.

“Cars with electrical drive systems represent a solution for the future, and will in steadily increasing degree be seen on the roads”.

However, this idea should soon prove largely to be both technologically and economically oversold. The setbacks were many. Before the turn of the century, the hopes, visions and car manufacturer interests had faded out. Now the hydrogen fuel cell electric car was more in focus. But this is another story. In the biofuel directive, the EU has quite realistically summarised the end of the last electric car wave in this way (EU COM, 2001):

“Electrical cars have been commercially available for several years, but have not been able to attract sufficient interest among consumers. The size and cost of batteries, relative to the energy quantity they carry, seem strongly limiting in order to produce a car of sufficient size, capacity and range between each recharging to a price buyers are willing to pay. In addition, the slow recharging of batteries, normally during night, is considered a drawback by potential buyers.

Expectations of breakthroughs in development of battery technologies, necessary to make the electrical car gain more appeal among larger customer segments, seem to be reduced the later years. Electrical cars may still have a niche market for transport purposes over shorter distances, where no noise and no emissions are essential. Unless a breakthrough in the development of battery technology is changing this scenario, the Commission sees small possibilities in keeping the electrical car on the list of candidates for alternative vehicles which can give markets of larger volumes”.

The long history of electric car development has not been a history of continuous innovation. On the contrary, it has been a history of many ups and downs. All major innovation steps were made more than a hundred years ago, regarding electrical motors, batteries, and recharging infrastructure. Even the speed and range records from this period are still quite impressive (Ulvönäs, 1983). In practice, the golden age has never returned, though both expectations and development investments have been substantial. The 1990s should never become more than a *gilded age*, an age already terminated. The ups have taken place in an ever-changing contexts—from local pollution and noise abatement to global sustainable development and climate change. But the ensuing downs—and the forms of *reverse salients*—have largely been the same; costly batteries, small ranges, slow speeds, and difficult and

time-consuming recharging conditions. The everlasting question on which type of power stations the electricity comes from has also been raised on a continuous basis. Learning from history may force us to ask whether these are necessary structural limitations, and whether the electric car can really become more than a niche vehicle for some limited urban use purposes. We are currently in the second era of hybrid car use. It is not a gilded era. Fully drivable models are on the market. But to what extent they will gain more permanence this time still needs to be demonstrated. This is not least the case of the plug-in adaptations.

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